

Guillermo A. Mendoza Franco

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Education

Pitzer College, Claremont, CA

B.A. in Neuroscience with Honors & Data Science

Graduation Date: May 2026

GPA: 3.80

Awards/Distinctions: Davis Scholar, Pitzer Scholar, Sigma Xi (Scientific Research Honor Society) Member

Relevant coursework: Data Structures & Algorithms, Machine Learning, Statistical Linear Models, Neural Networks, Discrete Math, Probability, Statistical Inference, Big Data, Applications of AI Engineering

Skills

Programming: Python, SQL, R, MATLAB, C++

Python/ML: Pandas, NumPy, SciPy, scikit-learn, Keras, data preprocessing, model validation, error analysis

Applied AI & Evaluation: Agentic RAG, LLM evaluation, grounded generation, retrieval analysis, source attribution, prompt-injection testing

Engineering Tools: Git, Linux, Docker, AWS Lambda, shell scripting, HTML/CSS/JavaScript, Excel, PowerBI, LaTeX

Projects

Grounded AI Resume Tailoring Agent

July 2026

- Built an agentic RAG application that extracts requirements, retrieves supporting evidence, and generates source-grounded recommendations, with automated checks for retrieval coverage, unsupported claims, and prompt injection
- Added source tracing, API logging, rate limiting, graceful error handling, and evaluation workflows to detect failures and improve system reliability

Los Angeles Parks RAG Guide

June 2026

- Built a modular RAG application over 664 chunks from 10 community, government, and travel sources, using source-specific chunking, persistent vector retrieval, grounded generation, and a Gradio interface
- Evaluated retrieval and generated responses across five query categories, reproduced a grounding failure, traced it to weak semantic retrieval, and designed a distance-threshold fallback to prevent unsupported outputs

Large-Scale Twitter Data Systems

February 2026

- Processed ~1.1B compressed geotagged tweets with a custom POSIX MapReduce pipeline, aggregating COVID-related hashtag usage across countries and languages without decompressing archives to disk
- Designed PostgreSQL indexes over approximately 31M records, reducing multi-minute scans to seconds by matching normalized and JSONB index strategies to observed query patterns

Work Experience

The Claremont Colleges

Claremont, CA

Portal & Form Developer

July 2025 – May 2026

- Designed and implemented student-facing digital forms and portal components using HTML/JavaScript, improving usability and structured data capture workflows
- Translated stakeholder needs into technical solutions by documenting requirements, workflows, and implementation decisions
- Collaborated with campus teams to test user workflows, diagnose usability and data-capture issues, and iteratively refine internal tools based on stakeholder feedback

AI Marketing Strategy Consultant — Airis Garden Practicum

January 2026 - May 2026

- Advised a Scandinavian plant-tech startup on LLM-native marketing strategy, translating ambiguous client needs and a D2C-to-B2B pivot into recommendations for AI discoverability, conversion, and chatbot deployment
- Analyzed Google Ads, sales, and market signals; built and presented an AI “Plant Coach” chatbot, communicating progress through weekly memos and final presentations

Research Assistant – beatLab

January 2024 – May 2026

- Built end-to-end reproducible Python pipelines for high-dimensional EEG data from preprocessing to model validation
- Compared models across subjects, improving predictive accuracy by 10 absolute percentage points over a baseline
- Completed an honors thesis on building reliable ML workflows for noisy, high-dimensional EEG data, communicating preprocessing, validation, error analysis, and baseline comparison results through code, figures, and a formal manuscript

Research Assistant & Lab Manager – Sheung Lab

May 2023 – August 2025

- Modeled emergent dynamics in active matter systems using MATLAB; analyzed 100+ hours of experimental data
- Managed research operations and trained 8+ assistants on protocols and reproducibility

Leadership & Outreach

Computer Science & Machine Learning Tutor – Pitzer College

August 2024 – May 2026

- Tutored peers in Python and ML coursework; led weekly theory and debugging sessions